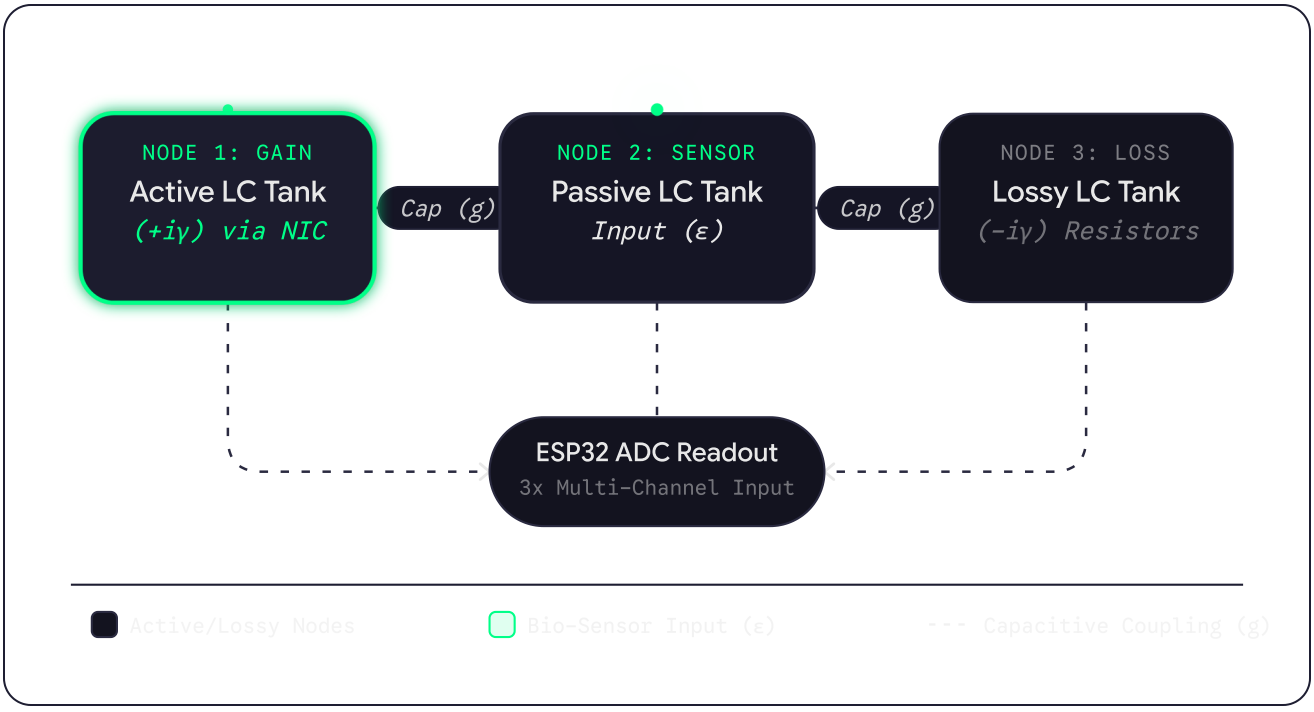
ATLAS • EP3 LAB • #020208 LIVE • 3-NODE

3-Resonator EP3 Circuit Architecture

Exceptional Point 3 (EP3) non-Hermitian sensor topology. Hover/select nodes to view matrix term. Bio-Sensor Input node exhibits cubic-root sensitivity $\Delta\omega \propto \varepsilon^{1/3}$.



Node 1: Active LC Tank (Gain)

$H[1,1] = \omega_o + i\gamma$

$$H_{EP3} = \begin{bmatrix} \omega_o + i\gamma & g & 0 \\ g & \omega_o + \varepsilon & g \\ 0 & g & \omega_o - i\gamma \end{bmatrix}$$

Active resonator with negative resistance implemented via Op-Amp Negative Impedance Converter (NIC). Supplies energy rate +iγ to balance system loss.

H[1,1]=Ω₀+IΓ • H[2,2]=Ω₀+E • H[3,3]=Ω₀-IΓ • G = CAP EP3 • ATLAS

EP3 SENSOR — INTERACTIVE VISUAL — BIO-SENSOR INPUT #020208 #e8e8e8 #00ff88